

# YAQIN CHEN

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## RESEARCH INTERESTS

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**Generative Modeling and its Foundations:** Deep Generative Models, Statistical Machine Learning, AI for Science

**Sequential Decision Making and Optimization:** Reinforcement Learning, Optimization Methods

## EDUCATION

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**School of Mathematics (Zhuhai), Sun Yat-sen University**

Sep 2022 – Jun 2026 (expected)

- B.S. in Information and Computing Science, GPA: 4.0/4.0 (1/60)

- Core curriculum: Mathematical Statistics, Real Analysis, Numerical Analysis, Stochastic Processes, Optimization, PDE

**Department of Statistics, University of Wisconsin-Madison**

Sep 2025 – May 2026 (expected)

- Visiting International Student Program (Thematic Statistics Track)

- Core curriculum: Statistical Inference, Statistical Methods, Machine Learning and Statistical Pattern

## RESEARCH EXPERIENCE

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**Decision, Risk and Operations Division, Columbia Business School** *Research Intern (remote)*

Mar 2025 – Present

Advisor: [Hongseok Namkoong](#)

- Developed the core scalable benchmarks for evaluating large language models in interactive, multi-turn environments.
- Benchmarked frontier LLM in-context learning across repeated interactions, identifying key adaptation challenges.
- Analyzed LLM failure modes via user simulation, providing key evidence of poor learning from experience.

**iSURE Program, University of Notre Dame** *Research Intern*

July 2025 – Oct 2025

Advisor: [Jianxun Wang](#) ([Sibley School of Mechanical Engineering, Cornell University](#))

- Researched the theoretical and mathematical unification of diffusion models and flow-matching generative frameworks.
- Adapted and benchmarked posterior sampling for flow-based conditional generation on scientific datasets (e.g., turbulence).
- Investigated training-free guidance techniques to accelerate sampling speed while ensuring generation accuracy.

**Human Cyber Physical Intelligence Laboratory, Sun Yat-sen University** *Research Intern*

Feb 2024 – Feb 2025

Advisor: [Pengxu Wei](#), [Liang Lin](#) ([Fellow of IEEE/IAPR](#))

- Researched the impact of latent alignment and structural inductive biases in Variational Autoencoders (VAEs).
- Developed a statistical framework for efficient representation learning with Minimum Variance Unbiased Estimator.
- Devised novel autoencoder architectures that enforce statistical efficiency to improve representation learning quality.

## ACADEMIC PAPERS

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**Benchmarking In-context Experiential Learning Through Repeated Product Recommendations**[\[pdf\]](#)

Gilbert Yang\*, **Yaqin Chen**\*, Thomson Yen\*, Hongseok Namkoong

*Under review at the International Conference on Learning Representations (ICLR), 2026*

**Training-Free Guided Generation techniques for Conditional Flow Matching on Scientific Datasets**(working paper)

Meet Hemant Parikh, **Yaqin Chen**, Jianxun Wang

**MVU-AE: Minimum Variance Unbiased Autoencoders**[\[pdf\]](#)

**Yaqin Chen**, Junrong Lian, Qing Zhao, Xiangyang Ji, Liang Lin, Pengxu Wei

## SELECTED PROJECTS

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**Development of a Martial Arts Teaching and Scoring System Based on YOLOv8**

[Demo Pages](#)

2024 Chinese Academy of Sciences Innovation and Practice Training Program For Undergraduate

Project Leader

- Developed a real-time martial arts coaching system using YOLOv8 for pose estimation and movement analysis.

- Engineered an automated scoring engine based on video analysis to provide objective performance feedback.
- Integrated a Large Language Model (LLM) to generate personalized, multi-modal (text and voice) coaching.

AWARDS

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|------------------------------------------------------------------------|------|
| Chinese National Scholarship (Top 1%)                                  | 2024 |
| First-Class Excellent Student of SYSU (Top 5%, 1/151)                  | 2024 |
| First-Class Excellent Student of SYSU (Top 5%, 1/151)                  | 2023 |
| MCM Modeling Contest Meritorious Award (7% of 11296)                   | 2023 |
| First Prize 37th Chinese Physics Olympiad of Jiangxi Province (Top 5%) | 2020 |

PROFESSIONAL SKILLS

**Programming Languages:** Python, C++ , R, MATLAB

**Languages:** English(fluent), Chinese(native)

STANDARDIZED TESTS

**GRE:** Verbal 159 (80%) + Quantitative 170 (92%) + Analytical Writing 3

**TOEFL:** Total 108 (Reading 28+Listening 27+Speaking 26+Writing 27)